

Practical 1: Microclimate modelling under different conditions

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1 Introduction

In this practical we will work with NicheMapR's microclimate model. The main aim is to simulate soil temperatures at a site in the Ecuadorian Andes (i.e., in Antisana).

Study site: Low elevation (-0.422961, -78.125341, ~3753 m)

In the next practical we will explore microclimatic differences across elevations, but this first practical session is focused on exploring the microclimate model's input and outputs thoroughly. These will include modifying shade levels and observe their effect, changing soil properties and see how they affect thermal conditions, and adding terrain (slope, aspect) and compare north vs south-facing slopes.

2 Setting up

First, load the NicheMapR package. It contains all the functions we need.

```
library(NicheMapR)
```

The global climate data should already be available. Let us verify:

```
gcfolder_file <- paste0(.libPaths()[1], "/gcfolder.rda")
if (file.exists(gcfolder_file)) {
  load(gcfolder_file)
  message("Global climate data is at: ", folder)
} else {
  message("Please run: get.global.climate(folder = 'path/to/data')")
}
```

```
## Global climate data is at: ~/Documents/gcfolder
```

3 Your first model run

The function `micro_global()` extracts climate data from a global database and runs the microclimate model. Let us run it for our low-elevation site with default settings.

```
micro <- micro_global(
  loc = c(-78.125341, -0.422961),
  elev = 3753,
  timeinterval = 12,
  nyears = 1
)
```

Note: `timeinterval = 12` runs the model for the average day of each month. `nyears = 1` runs a single year. The model first does a 3-iteration spin-up (using the output of day 1 as the input for day 2, and so on)

to stabilise the soil temperature profile.

The output object `micro` is a list containing matrices with the results:

- `micro$metout` — above-ground conditions under **minimum shade**
- `micro$shadmet` — above-ground conditions under **maximum shade**
- `micro$soil` — soil temperatures at 10 depths under minimum shade
- `micro$shadsoil` — soil temperatures at 10 depths under maximum shade

Let us look at the first few rows of `metout`:

```
head(micro$metout[, 1:10], 10)
```

```
##      DOY TIME      TALOC      TAREF
## [1,]  15    0  2.977710  1.652010
## [2,]  15   60  2.661546  1.471869
## [3,]  15  120  2.422471  1.332820
## [4,]  15  180  2.240032  1.225489
## [5,]  15  240  2.184444  1.142642
## [6,]  15  300  2.154265  1.078693
## [7,]  15  360  1.701019  0.978714
## [8,]  15  420  3.331806  2.218155
## [9,]  15  480  6.768828  4.608271
## [10,] 15  540 10.950599  7.725455
##      RHLOC      RH      VLOC
## [1,] 72.94244 80.17286 0.35136016
## [2,] 76.76480 83.57425 0.28954079
## [3,] 80.45225 86.97563 0.22772143
## [4,] 84.04174 90.37702 0.16590206
## [5,] 87.03089 93.77840 0.06739727
## [6,] 91.83549 99.19827 0.02189392
## [7,] 94.18035 99.19827 0.09184904
## [8,] 89.01036 96.33687 0.15241651
## [9,] 76.87074 89.29899 0.21768207
## [10,] 66.19848 82.26111 0.28055026
##      VREF SNOWMELT POOLDEP
## [1,] 1.60066286      0      0
## [2,] 1.31903740      0      0
## [3,] 1.03741194      0      0
## [4,] 0.75578649      0      0
## [5,] 0.30703624      0      0
## [6,] 0.09974033      0      0
## [7,] 0.44905898      0      0
## [8,] 0.79837762      0      0
## [9,] 1.14769626      0      0
## [10,] 1.49701491      0      0
```

The columns are:

Col	Name	Description
1	DOY	Day of year
2	TIME	Time of day (minutes)
3	TALOC	Air temperature at local height — °C
4	TAREF	Air temperature at reference height — °C
5	RHLOC	Relative humidity at local height — %
6	RH	Relative humidity at reference height — %

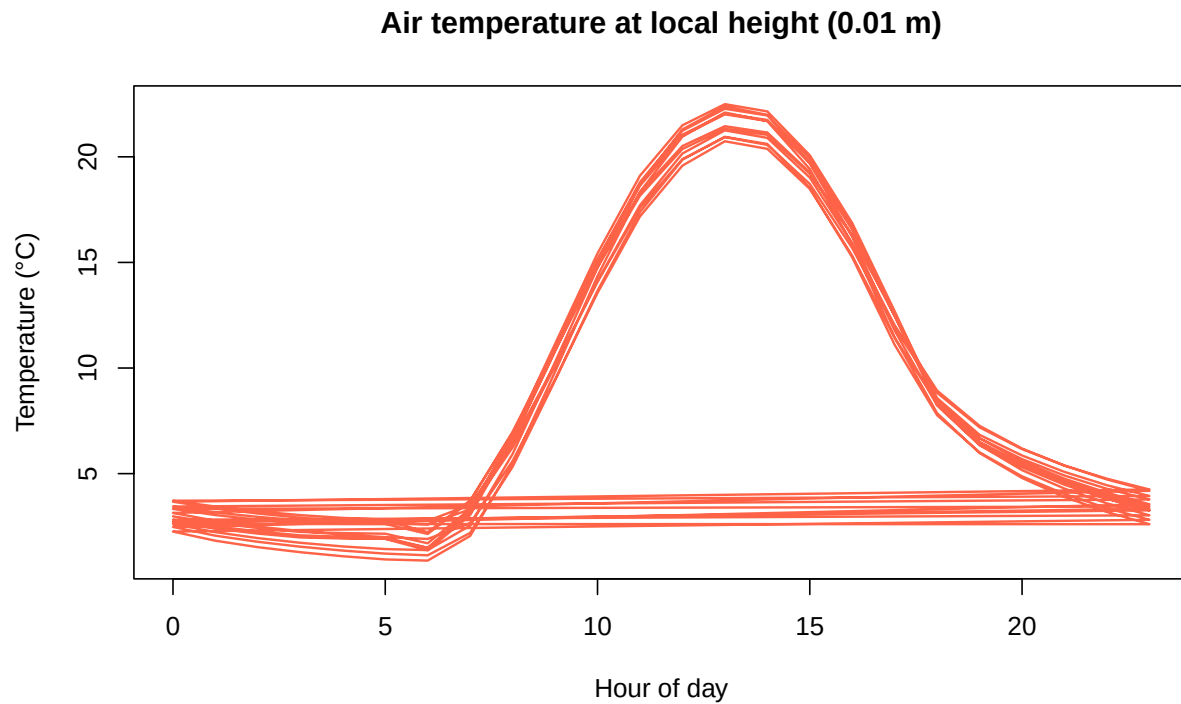
Col	Name	Description
7	VLOC	Wind speed at local height — m/s
8	VREF	Wind speed at reference height — m/s
12	ZEN	Solar zenith angle — degrees
13	SOLR	Solar radiation — W/m ²
14	TSKYC	Sky radiant temperature — °C

4 Plotting the results

Let us plot the hourly air temperature for each month, for both local and reference heights.

```
# Convert time to hours
metout <- cbind(micro$metout, hour = micro$metout[, 2] / 60)

# Plot air temperature at local height
plot(metout[, "hour"], metout[, "TALOC"],
     type = "l", col = "tomato", lwd = 1.5,
     xlab = "Hour of day", ylab = "Temperature (°C)",
     main = "Air temperature at local height (0.01 m)"
)
```



Now let us look at the soil temperature profiles under minimum shade vs maximum shade:

```
soil <- micro$soil
shadsoil <- micro$shadsoil

# Soil depths
depths <- c("D0cm", "D2.5cm", "D5cm", "D10cm", "D15cm",
```

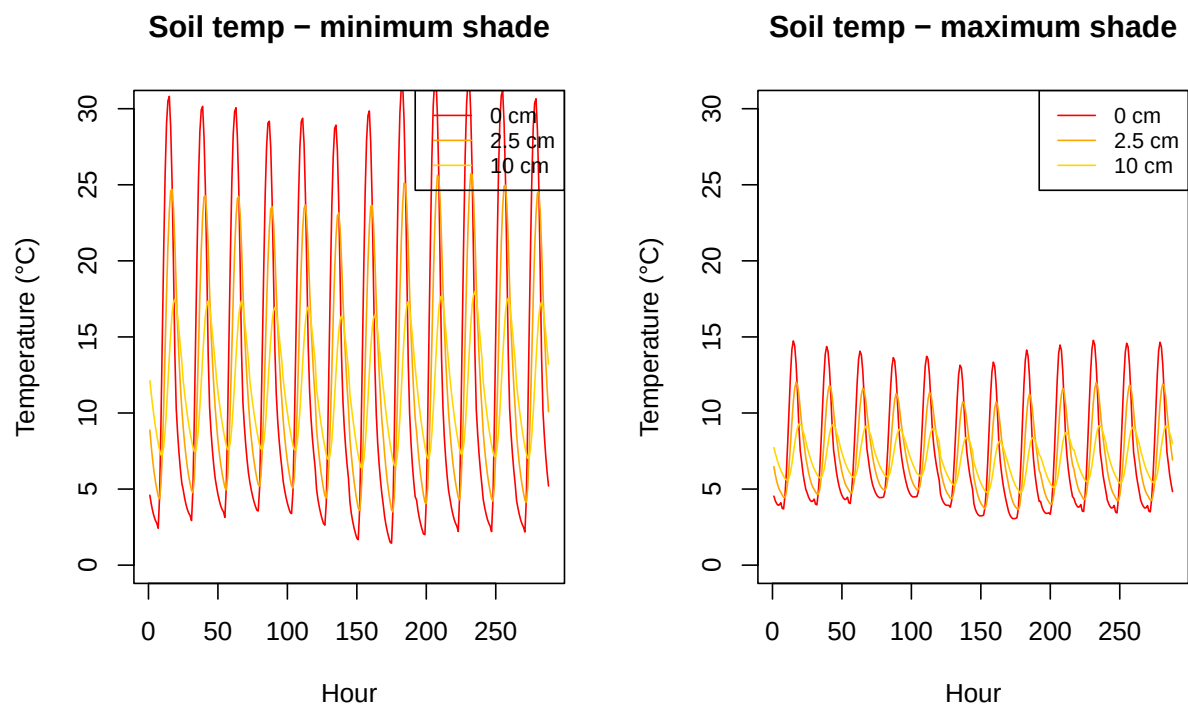
```

"D20cm", "D30cm", "D50cm", "D100cm", "D200cm")

# Plot soil temperature at 0 cm, 2.5 cm, and 10 cm under min shade
par(mfrow = c(1, 2))
plot(soil[, "D0cm"], type = "l", col = "red", ylim = c(0, 30),
     xlab = "Hour", ylab = "Temperature (°C)",
     main = "Soil temp - minimum shade")
lines(soil[, "D2.5cm"], col = "orange")
lines(soil[, "D10cm"], col = "gold")
legend("topright", legend = c("0 cm", "2.5 cm", "10 cm"),
     col = c("red", "orange", "gold"), lty = 1, cex = 0.8)

plot(shadsoil[, "D0cm"], type = "l", col = "red", ylim = c(0, 30),
     xlab = "Hour", ylab = "Temperature (°C)",
     main = "Soil temp - maximum shade")
lines(shadsoil[, "D2.5cm"], col = "orange")
lines(shadsoil[, "D10cm"], col = "gold")
legend("topright", legend = c("0 cm", "2.5 cm", "10 cm"),
     col = c("red", "orange", "gold"), lty = 1, cex = 0.8)

```



What do you notice? The maximum shade scenario keeps soil temperatures cooler during the day but slightly warmer at night (due to longwave radiation from vegetation). The deeper layers (10 cm) show less daily variation.

5 Shade effects

Different shade levels simulate different habitat types: open soil, partial canopy, and dense forest. Let us run the model with three distinct shade levels.

```

# Open: no shade
micro_open <- micro_global(
  loc = c(-78.125341, -0.422961),
  elev = 3753,
  timeinterval = 12, runshade = 0, minshade = 0
)

# Partial shade (50%)
micro_partial <- micro_global(
  loc = c(-78.125341, -0.422961),
  elev = 3753,
  timeinterval = 12, runshade = 0, minshade = 50
)

# Dense shade (90%)
micro_dense <- micro_global(
  loc = c(-78.125341, -0.422961),
  elev = 3753,
  timeinterval = 12, minshade = 90, maxshade = 100
)

```

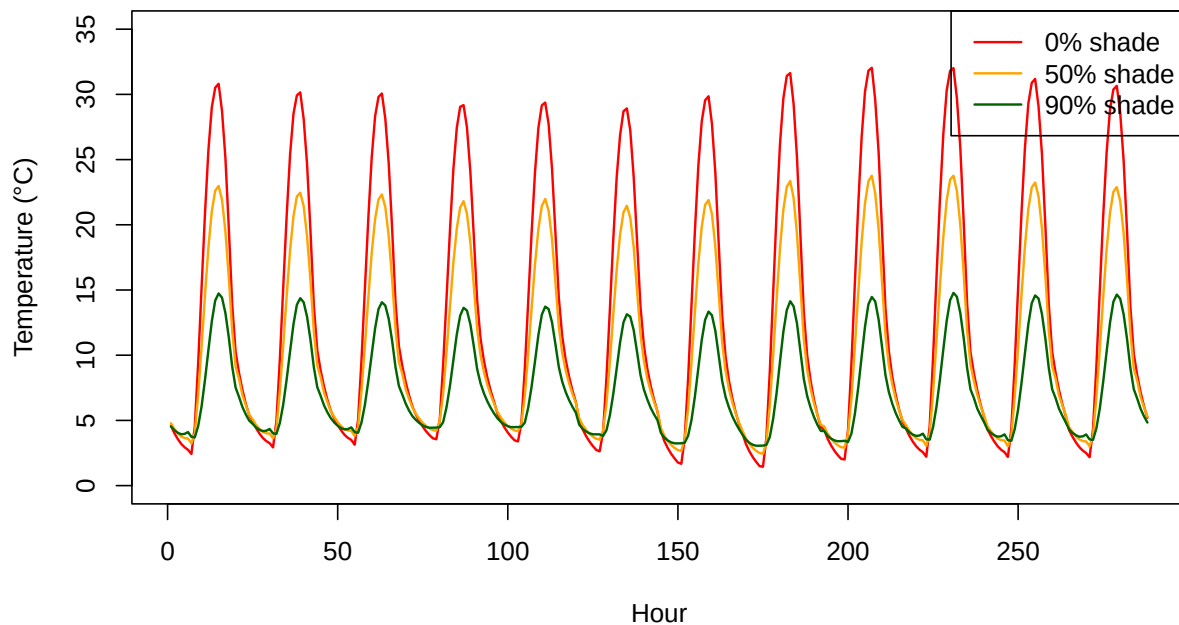
Now compare the soil surface temperatures:

```

par(mfrow = c(1, 1))
plot(micro_open$soil[, "D0cm"], type = "l", col = "red", lwd = 1.5,
  ylim = c(0, 35), xlab = "Hour", ylab = "Temperature (°C)",
  main = "Soil surface temperature under different shade levels")
lines(micro_partial$soil[, "D0cm"], col = "orange", lwd = 1.5)
lines(micro_dense$soil[, "D0cm"], col = "darkgreen", lwd = 1.5)
legend("topright", legend = c("0% shade", "50% shade", "90% shade"),
  col = c("red", "orange", "darkgreen"), lty = 1, lwd = 1.5)

```

Soil surface temperature under different shade levels



What do you observe? Dense shade reduces peak temperatures by 10–15°C. This has huge implications for egg incubation — shaded nests are much cooler.

5.1 Challenge 1: Optimal incubation temperature

Many tropical reptile eggs incubate best between 28–32°C. Run the model with `runshade = 0`, `minshade = 0`, 25, 50, 75, 90 and find which shade level keeps the soil temperature at 5 cm depth closest to 30°C during the hottest part of the day.

Hint: extract `micro$soil[, "D5cm"]` for each run and compute the mean or median.

```
# Your code here
shade_levels <- c(0, 25, 50, 75, 90)
# Fill in the loop to run the model for each shade level
# and extract D5cm temperatures
```

6 Soil properties

Soil type strongly influences thermal conductivity and heat capacity. NicheMapR provides soil types 0 (rock) through 11 (clay), plus the option to specify custom properties.

Let us compare sand (`soiltype = 1`), loam (`soiltype = 4`), and rock (`soiltype = 0`):

```
micro_sand <- micro_global(
  loc = c(-78.125341, -0.422961),
  elev = 3753,
  timeinterval = 12, soiltype = 1
)
micro_loam <- micro_global(
  loc = c(-78.125341, -0.422961),
```

```

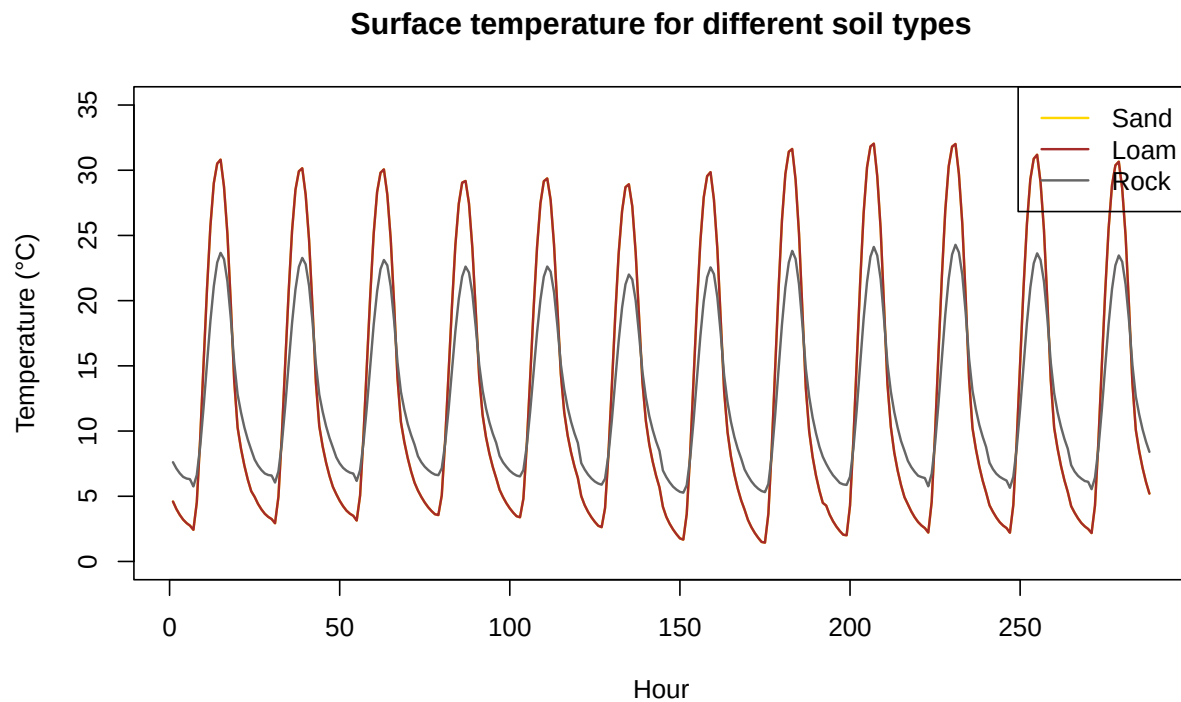
    elev = 3753,
    timeinterval = 12, soiltype = 4
)
micro_rock <- micro_global(
  loc = c(-78.125341, -0.422961),
  elev = 3753,
  timeinterval = 12, soiltype = 0
)

```

```

plot(micro_sand$soil[, "D0cm"], type = "l", col = "gold", lwd = 1.5,
     ylim = c(0, 35), xlab = "Hour", ylab = "Temperature (°C)",
     main = "Surface temperature for different soil types")
lines(micro_loam$soil[, "D0cm"], col = "brown", lwd = 1.5)
lines(micro_rock$soil[, "D0cm"], col = "gray40", lwd = 1.5)
legend("topright", legend = c("Sand", "Loam", "Rock"),
     col = c("gold", "brown", "gray40"), lty = 1, lwd = 1.5)

```



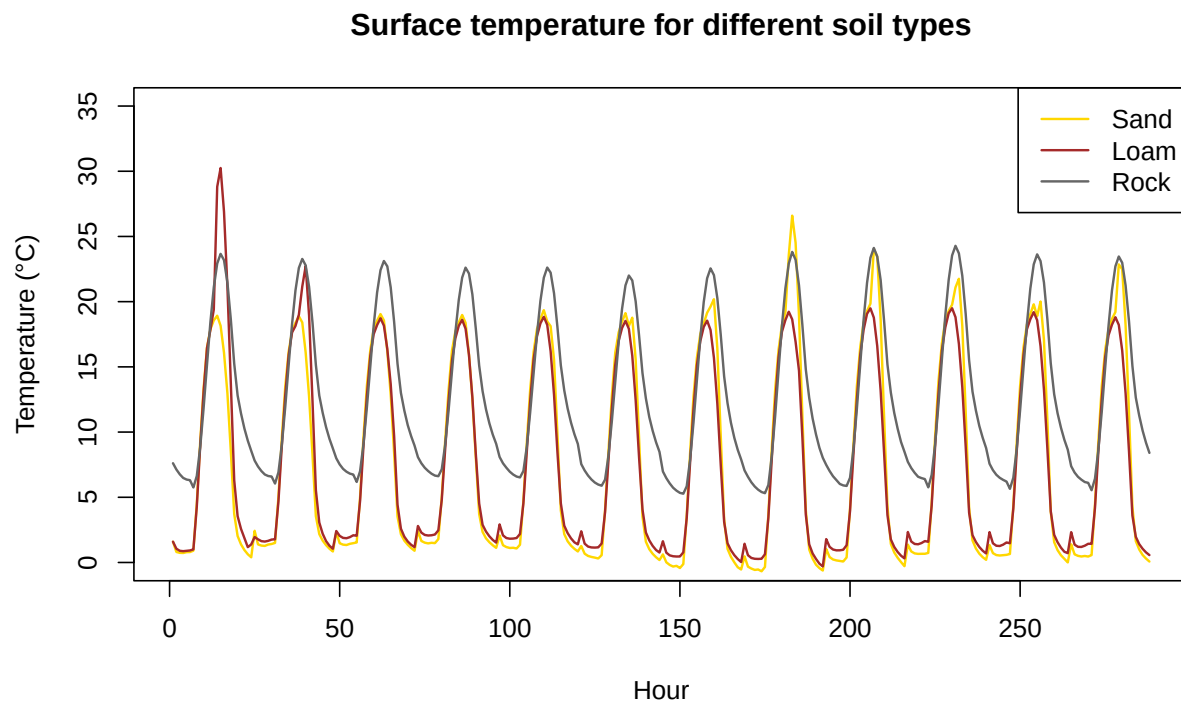
```

micro_sand <- micro_global(
  loc = c(-78.125341, -0.422961),
  elev = 3753,
  runmoist = 1,
  timeinterval = 12, soiltype = 1
)
micro_loam <- micro_global(
  loc = c(-78.125341, -0.422961),
  elev = 3753,
  runmoist = 1,
  timeinterval = 12, soiltype = 4
)

```

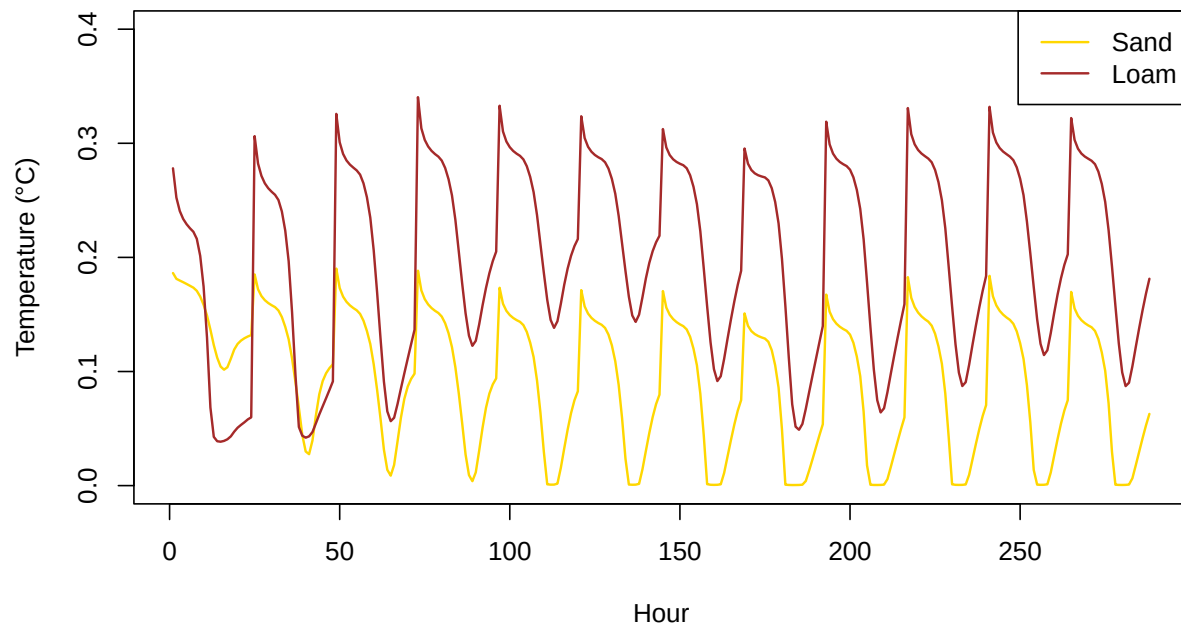
```
micro_rock <- micro_global(
  loc = c(-78.125341, -0.422961),
  elev = 3753,
  runmoist = 1,
  timeinterval = 12, soiltype = 0
)
```

```
plot(micro_sand$soil[, "D0cm"], type = "l", col = "gold", lwd = 1.5,
  ylim = c(0, 35), xlab = "Hour", ylab = "Temperature (°C)",
  main = "Surface temperature for different soil types")
lines(micro_loam$soil[, "D0cm"], col = "brown", lwd = 1.5)
lines(micro_rock$soil[, "D0cm"], col = "gray40", lwd = 1.5)
legend("topright", legend = c("Sand", "Loam", "Rock"),
  col = c("gold", "brown", "gray40"), lty = 1, lwd = 1.5)
```



```
plot(micro_sand$soilmoist[, "WC0cm"], type = "l", col = "gold", lwd = 1.5,
  ylim = c(0, 0.4), xlab = "Hour", ylab = "Temperature (°C)",
  main = "Surface temperature for different soil types")
lines(micro_loam$soilmoist[, "WC0cm"], col = "brown", lwd = 1.5)
# lines(micro_rock$soilmoist[, "WC0cm"], col = "gray40", lwd = 1.5)
legend("topright", legend = c("Sand", "Loam"),
  col = c("gold", "brown"), lty = 1, lwd = 1.5)
```


Surface temperature for different soil types

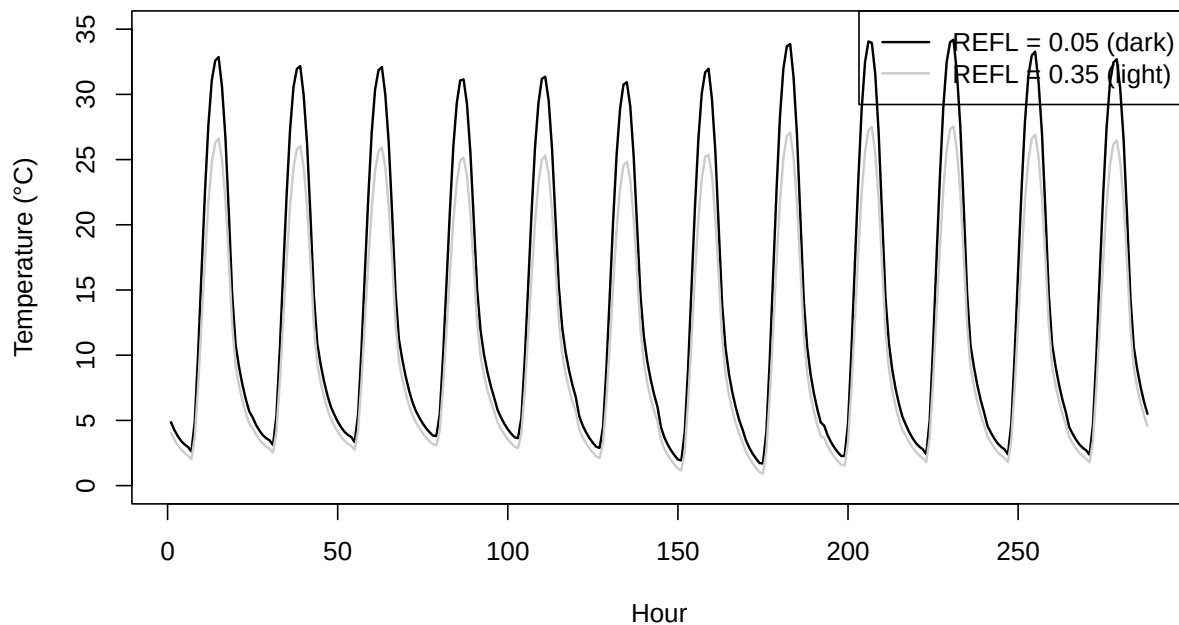


You can also change the soil solar reflectance (REFL). This is the albedo:

```
micro_dark <- micro_global(
  loc = c(-78.125341, -0.422961),
  elev = 3753,
  timeinterval = 12, REFL = 0.05 # dark soil
)
micro_light <- micro_global(
  loc = c(-78.125341, -0.422961),
  elev = 3753,
  timeinterval = 12, REFL = 0.35 # light soil
)

plot(micro_dark$soil[, "D0cm"], type = "l", col = "black", lwd = 1.5,
  ylim = c(0, 35), xlab = "Hour", ylab = "Temperature (°C)",
  main = "Effect of soil reflectance on surface temperature")
lines(micro_light$soil[, "D0cm"], col = "gray80", lwd = 1.5)
legend("topright", legend = c("REFL = 0.05 (dark)", "REFL = 0.35 (light)"),
  col = c("black", "gray80"), lty = 1, lwd = 1.5)
```

Effect of soil reflectance on surface temperature



Darker soils absorb more solar radiation and reach higher temperatures. This is why tilled fields can be much hotter than bare rock of the same colour.

6.1 Challenge 2: Daily temperature range by soil type

Compare the daily temperature range (maximum minus minimum at the surface) for sand, loam, and rock. Which soil type shows the largest range? Why?

Hint: use `max()` and `min()` on the surface temperature column.

Your code here

7 Terrain effects

Slope and aspect change the amount of solar radiation received. In the Southern Hemisphere, north-facing slopes receive more sun (opposite to the Northern Hemisphere).

Compare a flat surface, a steep north-facing slope, and a steep south-facing slope. Remember that Ecuador is in the Southern Hemisphere, so `aspect = 0` (north) receives more direct sunlight.

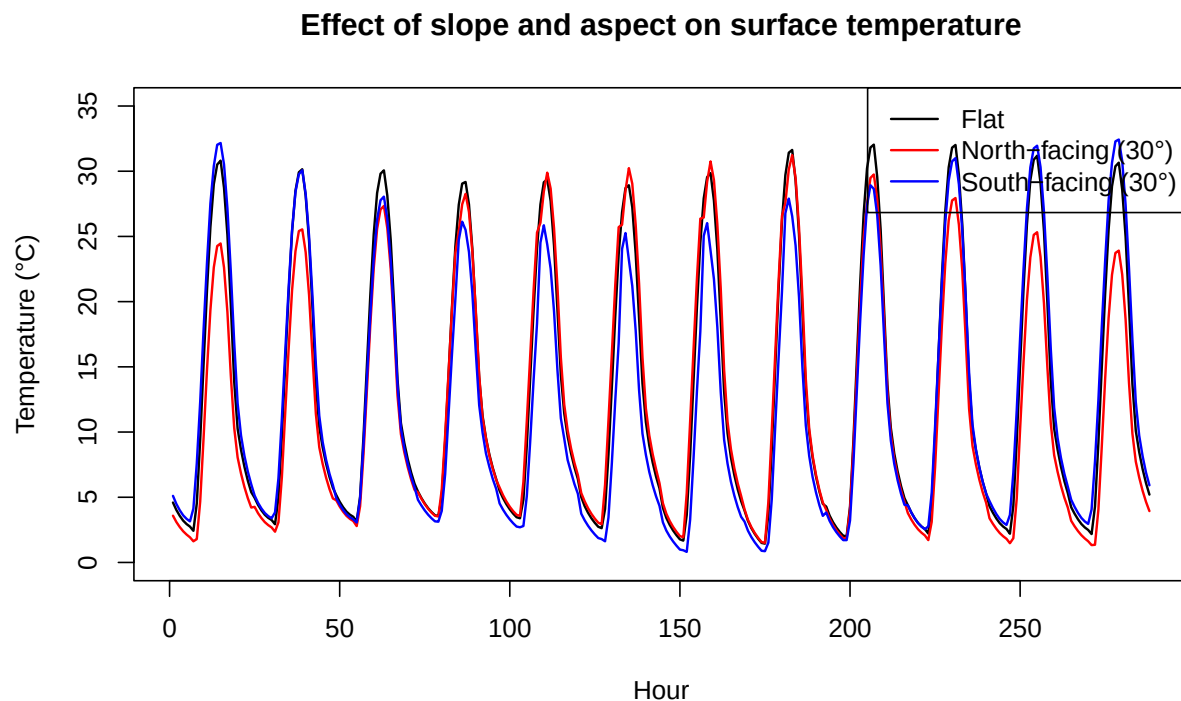
```
micro_flat <- micro_global(  
  loc = c(-78.125341, -0.422961),  
  elev = 3753,  
  timeinterval = 12, slope = 0, aspect = 0  
)  
micro_north <- micro_global(  
  loc = c(-78.125341, -0.422961),  
  elev = 3753,  
  timeinterval = 12, slope = 30, aspect = 0  
)
```

```

micro_south <- micro_global(
  loc = c(-78.125341, -0.422961),
  elev = 3753,
  timeinterval = 12, slope = 30, aspect = 180
)

par(mfrow = c(1, 1))
plot(micro_flat$soil[, "D0cm"], type = "l", col = "black", lwd = 1.5,
  ylim = c(0, 35), xlab = "Hour", ylab = "Temperature (°C)",
  main = "Effect of slope and aspect on surface temperature")
lines(micro_north$soil[, "D0cm"], col = "red", lwd = 1.5)
lines(micro_south$soil[, "D0cm"], col = "blue", lwd = 1.5)
legend("topright", legend = c("Flat", "North-facing (30°)", "South-facing (30°)"),
  col = c("black", "red", "blue"), lty = 1, lwd = 1.5)

```



Notice the difference: In the Southern Hemisphere, north-facing slopes receive more solar energy, so they are substantially hotter. South-facing slopes can be 5–10°C cooler — a huge difference for organisms.

7.1 Challenge 3: Slope aspect and nesting

Imagine you are a reptile choosing a nest site on a 20° slope. Compare `aspect = 0` (north), `aspect = 90` (east), `aspect = 180` (south), and `aspect = 270` (west). Which aspect gives the warmest incubation conditions at 2.5 cm depth? Which gives the coolest?

```

# Your code here
aspects <- c(0, 90, 180, 270)

```

8 Summary

In this practical you have learned to:

1. Run NicheMapR's `micro_global()` function
2. Extract and plot air temperature and soil temperature outputs
3. Modify shade levels and understand their effect on ground temperatures
4. Change soil types and soil reflectance
5. Add slope and aspect, and compare thermal conditions between orientations